CSE440: Natural Language Processing II

Lab Assignment 1

1. Write a Python program using NLTK to explore two corpora. First, load the Reuters corpus and display all of its available categories using appropriate NLTK functions. Then, load the Movie Reviews corpus and display the list of available text files.

2. Select two text files from the State of the Union corpus. Tokenize both texts into words and generate separate word clouds for each to compare the most frequent words used in both files.

3. Select a text from the Movie Reviews corpus that belongs to the 'neg' (negative) category. Preprocess it by tokenizing into words, removing stopwords, and applying lemmatization. Identify the 20 most frequent words in the cleaned text and visualize their frequencies using a bar plot.

4. Choose a text file from the Reuters corpus in NLTK. Preprocess the text by converting it to lowercase, removing punctuation, and filtering out stopwords. Identify the top 12 most frequent words from the cleaned text. Then, use bigrams (2-grams) to extract all consecutive word pairs. Build a co-occurrence matrix showing how often each pair of the top 12 words appears together as a bigram in the text.

5. Write a Python program that processes the German text from the udhr corpus in NLTK. For each word in the text, extract the sequence of vowels it contains (ignore consonants). From these vowel sequences, generate all possible trigrams of consecutive vowels. Build a frequency distribution of these vowel trigrams across the entire text and display the result as a sorted list showing each trigram and its count.